

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5303C

5 Credits: 4

Programme Elective

Source-grounded

Embedded and IOT System Design

- To understand the basic concepts and applications of embedded system design

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5303C is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5303C.pdf from the uploaded official SITTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Robotic Process Automation
Course code	5303C
Course title	Embedded and IOT System Design
Semester	5 Credits: 4
Course category	Programme Elective
Periods per week	4 (L: 4 T: 0 P: 0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

34 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To understand the basic concepts and applications of embedded system design
- To understand the concepts of programming an embedded system and basics of Microcontroller
- To understand the fundamentals of IoT and its architectural modelling
- To understand and have thorough knowledge about smart things, functional blocks,
- communication networks and protocols used in IoT

Course outcomes

CO	Official outcome	Study level
CO1	14 Understanding structure and design of an Embedded System To understand the concepts of programming an	See official syllabus
CO2	15 Understanding embedded system and basics of Microcontroller To Understand the IoT fundamentals and architecture	See official syllabus
CO3	15 Understanding modeling To understand the smart things, functional blocks,	See official syllabus
CO4	14 Understanding communication networks and protocols used in IoT	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3 3
CO3	CO3 3 3 2

Row	Extracted official mapping
CO4	CO4 3 1

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Embedded System	7	As specified
Module 2	of a Microcontroller	8	As specified
Module 3	To Understand the IoT fundamentals and architecture modeling	10	As specified
Module 4	and protocols used in IoT	9	As specified

MODULE 1

Embedded System

This module is presented from the official syllabus scope for Embedded and IOT System Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Embedded System
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

▼ History, Embedded Systems vs general computing 1 Knowledge systems Describe the Characteristics and quality attributes of

History, Embedded Systems vs general computing 1 Knowledge systems Describe the Characteristics and quality attributes of is an official topic in Module 1 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify History, Embedded Systems vs general computing 1 Knowledge systems Describe the Characteristics and quality attributes of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, history, embedded systems vs general computing 1 knowledge systems describe the characteristics and quality attributes of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What History, Embedded Systems vs general computing 1 Knowledge systems Describe the Characteristics and quality attributes of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എം History, Embedded Systems vs general computing 1 Knowledge systems Describe the Characteristics and quality attributes of ഏതെങ്കിലും ഏതെങ്കിലും definition/meaning ഏതെങ്കിലും. ഏതെങ്കിലും ഏതെങ്കിലും steps, application ഏതെങ്കിലും ഏതെങ്കിലും. Technical terms English-എം ഏതെങ്കിലും ഏതെങ്കിലും.

▼ 2 Understanding Embedded systems,

2 Understanding Embedded systems, is an official topic in Module 1 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Embedded systems, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding embedded systems, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Embedded systems, means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding Embedded systems,
 ഫലപ്രസാദം definition/meaning
 ഫലപ്രസാദം, steps, application
 English-
 Technical terms

▼ Explain the Classification of Embedded systems 1 Understanding Describe about the Core of a typical embedded

Explain the Classification of Embedded systems 1 Understanding Describe about the Core of a typical embedded is an official topic in Module 1 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the Classification of Embedded systems 1 Understanding Describe about the Core of a typical embedded using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the classification of embedded systems 1 understanding describe about the core of a typical embedded should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the Classification of Embedded systems 1 Understanding Describe about the Core of a typical embedded means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain the Classification of Embedded systems 1 Understanding Describe about the Core of a typical embedded definition/meaning , steps, application Technical terms English-

▼ systems, Memory, Sensor and actuators, 3 Understanding Communication Interface, Embedded firmware Explain the Major application areas of embedded

systems, Memory, Sensor and actuators, 3 Understanding Communication Interface, Embedded firmware Explain the Major application areas of embedded is an official topic in Module 1 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify systems, Memory, Sensor and actuators, 3 Understanding Communication Interface, Embedded firmware Explain the Major application areas of embedded using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, systems, memory, sensor and actuators, 3 understanding communication interface, embedded firmware explain the major application areas of embedded should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What systems, Memory, Sensor and actuators, 3 Understanding Communication Interface, Embedded firmware Explain the Major application areas of embedded means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എം സിസ്റ്റം, മെമറി, സെൻസർ ആക്ടുവേറ്റർ, 3 Understanding Communication Interface, Embedded firmware Explain the Major application areas of embedded
 definition/meaning
 .
 , steps, application
 . Technical terms English-
 .

▼ 1 Understanding systems Describe about the Design Process in Embedded

1 Understanding systems Describe about the Design Process in Embedded is an official topic in Module 1 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding systems Describe about the Design Process in Embedded using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding systems describe about the design process in embedded should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding systems Describe about the Design Process in Embedded systems definition/meaning, steps, application, technical terms English- Malayalam.

▼ System, Design Metrics, Challenges in Embedded 3 Understanding System Design, Explain about Design Process and Design Example of

System, Design Metrics, Challenges in Embedded 3 Understanding System Design, Explain about Design Process and Design Example of is an official topic in Module 1 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify System, Design Metrics, Challenges in Embedded 3 Understanding System Design, Explain about Design Process and Design Example of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, system, design metrics, challenges in embedded 3 understanding system design, explain about design process and design example of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What System, Design Metrics, Challenges in Embedded 3 Understanding System Design, Explain about Design Process and Design Example of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-System, Design Metrics, Challenges in Embedded 3 Understanding System Design, Explain about Design Process and Design Example of definition/meaning. steps, application. Technical terms English-.

▼ 3 Applying Smart Card Introduction to Embedded Systems: Definition, History, Embedded Systems vs general computing systems, Characteristics and quality attributes of Embedded systems, Classification of Embedded systems, Core of a typical embedded systems, Memory, Sensor and actuators, Communication Interface, Embedded firmware, Major application areas of embedded systems, Design Process in Embedded System, Design Metrics, Challenges in Embedded System Design, Design Process and Design Example of Smart Card To understand the concepts of programming an embedded system and basics

3 Applying Smart Card Introduction to Embedded Systems: Definition, History, Embedded Systems vs general computing systems, Characteristics and quality attributes of Embedded systems, Classification of Embedded systems, Core of a typical embedded systems, Memory, Sensor and actuators, Communication Interface, Embedded firmware, Major application areas of embedded systems, Design Process in Embedded System, Design Metrics, Challenges in Embedded System Design, Design Process and Design Example of Smart Card To understand the concepts of programming an embedded system and basics is an official topic in Module 1 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying Smart Card Introduction to Embedded Systems: Definition, History, Embedded Systems vs general computing

systems, Characteristics and quality attributes of Embedded systems, Classification of Embedded systems, Core of a typical embedded systems, Memory, Sensor and actuators, Communication Interface, Embedded firmware, Major application areas of embedded systems, Design Process in Embedded System, Design Metrics, Challenges in Embedded System Design, Design Process and Design Example of Smart Card To understand the concepts of programming an embedded system and basics using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying smart card introduction to embedded systems: definition, history, embedded systems vs general computing systems, characteristics and quality attributes of embedded systems, classification of embedded systems, core of a typical embedded systems, memory, sensor and actuators, communication interface, embedded firmware, major application areas of embedded systems, design process in embedded system, design metrics, challenges in embedded system design, design process and design example of smart card to understand the concepts of programming an embedded system and basics should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying Smart Card Introduction to Embedded Systems: Definition, History, Embedded Systems vs general computing systems, Characteristics and quality attributes of Embedded systems, Classification of Embedded systems, Core of a typical embedded systems, Memory, Sensor and actuators, Communication Interface, Embedded firmware, Major application areas of embedded systems, Design Process in Embedded System, Design Metrics, Challenges in

Aspect	What to prepare
	Embedded System Design, Design Process and Design Example of Smart Card To understand the concepts of programming an embedded system and basics means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Applying Smart Card Introduction to Embedded Systems: Definition, History, Embedded Systems vs general computing systems, Characteristics and quality attributes of Embedded systems, Classification of Embedded systems, Core of a typical embedded systems, Memory, Sensor and actuators, Communication Interface, Embedded firmware, Major application areas of embedded systems, Design Process in Embedded System, Design Metrics, Challenges in Embedded System Design, Design Process and Design Example of Smart Card To understand the concepts of programming an embedded system and basics
 പരമ്പരാഗതമായി നിർവ്വചിക്കപ്പെട്ട definition/meaning ഉൾക്കൊള്ളുന്നു. പരമ്പരാഗതമായി നിർവ്വചിക്കപ്പെട്ട, steps, application ഉൾക്കൊള്ളുന്നു. Technical terms English-ൽ ഉൾക്കൊള്ളുന്നു.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2**of a Microcontroller**

This module is presented from the official syllabus scope for Embedded and IOT System Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: of a Microcontroller
- Official duration: As specified
- Official topic entries captured: 8
- The full source excerpt is preserved below for coverage checking.

▼ Programming Embedded Systems- Processor types 1 Understanding. Explain about Selection of programming language,

Programming Embedded Systems- Processor types 1 Understanding. Explain about Selection of programming language, is an official topic in Module 2 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Programming Embedded Systems- Processor types 1 Understanding. Explain about Selection of programming language, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, programming embedded systems- processor types 1 understanding. explain about selection of programming language, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Programming Embedded Systems- Processor types 1 Understanding. Explain about Selection of programming language, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്നുൾപ്പെടെ Programming Embedded Systems- Processor types 1 Understanding. Explain about Selection of programming language, ഏതെങ്കിലും ഒരു definition/meaning നൽകുക. ഏതെങ്കിലും ഏതെങ്കിലും steps, application നൽകുക. Technical terms English-ൽ നൽകുക.

▼ 2 Understanding Selection of Operating System Describe about the Development of Embedded System

2 Understanding Selection of Operating System Describe about the Development of Embedded System is an official topic in Module 2 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Selection of Operating System Describe about the Development of Embedded System using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding selection of operating system describe about the development of embedded system should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Selection of Operating System Describe about the Development of Embedded System means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Understanding Selection of Operating System Describe about the Development of Embedded System
 പരീക്ഷയിൽ definition/meaning നൽകണം. പരീക്ഷയിൽ പരീക്ഷണ
 പരീക്ഷണ, steps, application നൽകണം പരീക്ഷണ പരീക്ഷണ. Technical terms
 English-ൽ പരീക്ഷ പരീക്ഷണ.

▼ 2 Understanding. Software Block diagram of 8051, Pin configuration, Registers,

2 Understanding. Software Block diagram of 8051, Pin configuration, Registers, is an official topic in Module 2 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. Software Block diagram of 8051, Pin configuration, Registers, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. software block diagram of 8051, pin configuration, registers, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Understanding. Software Block diagram of 8051, Pin configuration, Registers, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Understanding. Software Block diagram of 8051, Pin configuration, Registers, definition/meaning. , steps, application . Technical terms English- .

▼ 2 Understanding. Internal Memory, Timers: Modes and Applications, Port Structures,

2 Understanding. Internal Memory, Timers: Modes and Applications, Port Structures, is an official topic in Module 2 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding. Internal Memory, Timers: Modes and Applications, Port Structures, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding. internal memory, timers: modes and applications, port structures, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What 2 Understanding. Internal Memory, Timers: Modes and Applications, Port Structures, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Understanding. Internal Memory, Timers: Modes and Applications, Port Structures, definition/meaning . . . , steps, application Technical terms English-

▼ 2 Applying Interrupts Assembly Language Programming - Addressing Modes,

2 Applying Interrupts Assembly Language Programming - Addressing Modes, is an official topic in Module 2 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying Interrupts Assembly Language Programming - Addressing Modes, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying interrupts assembly language programming - addressing modes, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Applying Interrupts Assembly Language Programming - Addressing Modes, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Applying Interrupts Assembly Language Programming - Addressing Modes, definition/meaning. steps, application. Technical terms English- .

▼ 2 Applying Instruction set Interfacing with 8051 using Assembly

2 Applying Instruction set Interfacing with 8051 using Assembly is an official topic in Module 2 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying Instruction set Interfacing with 8051 using Assembly using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying instruction set interfacing with 8051 using assembly should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Applying Instruction set Interfacing with 8051 using Assembly means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Applying Instruction set Interfacing with 8051 using Assembly ഏതുതരത്തിലുള്ള definition/meaning ഉണ്ടാകുന്നു. ഏതുതരത്തിൽ steps, application ഉണ്ടാകുന്നു. Technical terms English-ൽ ഉണ്ടാകുന്നു.

▼ language programming: LED, Seven segment LED 3 Applying display. Interfacing with 8051 using Assembly

language programming: LED, Seven segment LED 3 Applying display. Interfacing with 8051 using Assembly is an official topic in Module 2 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify language programming: LED, Seven segment LED 3 Applying display. Interfacing with 8051 using Assembly using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, language programming: led, seven segment led 3 applying display. interfacing with 8051 using assembly should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What language programming: LED, Seven segment LED 3 Applying display. Interfacing with 8051 using Assembly means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- language programming: LED, Seven segment LED 3 Applying display. Interfacing with 8051 using Assembly definition/meaning . steps, application . Technical terms English- .

▼ language programming: LCD display, Keyboard, 1 Applying Stepper Motor Programming Embedded Systems- Processor types, Selection of programming language, Selection of Operating System, Development of Embedded System Software. 8051 Microcontroller- Block diagram of 8051, Pin configuration, Registers, Internal Memory, Timers: Modes and Applications, Port Structures, Assembly Language Programming - Addressing Modes, Instruction set, Interrupts, Interfacing with 8051 using Assembly language programming: LED, Seven segment LED display. LCD display, Keyboard, Stepper Motor

language programming: LCD display, Keyboard, 1 Applying Stepper Motor Programming Embedded Systems- Processor types, Selection of programming language, Selection of Operating System, Development of Embedded System Software. 8051 Microcontroller- Block diagram of 8051, Pin configuration, Registers, Internal Memory, Timers: Modes and Applications, Port Structures, Assembly Language Programming - Addressing Modes, Instruction set, Interrupts, Interfacing with 8051 using Assembly language programming: LED, Seven segment LED display. LCD display, Keyboard, Stepper Motor is an official topic in Module 2 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify language programming: LCD display, Keyboard, 1 Applying Stepper Motor Programming Embedded Systems- Processor types, Selection of programming language, Selection of Operating System,

Development of Embedded System Software. 8051 Microcontroller- Block diagram of 8051, Pin configuration, Registers, Internal Memory, Timers: Modes and Applications, Port Structures, Assembly Language Programming - Addressing Modes, Instruction set, Interrupts, Interfacing with 8051 using Assembly language programming: LED, Seven segment LED display. LCD display, Keyboard, Stepper Motor using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, language programming: Lcd display, keyboard, 1 applying stepper motor programming embedded systems- processor types, selection of programming language, selection of operating system, development of embedded system software. 8051 microcontroller- block diagram of 8051, pin configuration, registers, internal memory, timers: modes and applications, port structures, assembly language programming - addressing modes, instruction set, interrupts, interfacing with 8051 using assembly language programming: led, seven segment led display. Lcd display, keyboard, stepper motor should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What language programming: LCD display, Keyboard, 1 Applying Stepper Motor Programming Embedded Systems- Processor types, Selection of programming language, Selection of Operating System, Development of Embedded System Software. 8051 Microcontroller- Block diagram of 8051, Pin configuration, Registers, Internal Memory, Timers: Modes and Applications, Port Structures, Assembly Language Programming - Addressing Modes, Instruction set, Interrupts, Interfacing with 8051 using Assembly language programming: LED, Seven segment LED display. LCD display, Keyboard, Stepper Motor means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- language programming: LCD display, Keyboard, 1 Applying Stepper Motor Programming Embedded Systems- Processor types, Selection of programming language, Selection of Operating System, Development of Embedded System Software. 8051 Microcontroller- Block diagram of 8051, Pin configuration, Registers, Internal Memory, Timers: Modes and Applications, Port Structures, Assembly Language Programming - Addressing Modes, Instruction set, Interrupts, Interfacing with 8051 using Assembly language programming: LED, Seven segment LED display. LCD display, Keyboard, Stepper Motor definition/meaning . steps, application . Technical terms English- .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

To Understand the IoT fundamentals and architecture modeling

This module is presented from the official syllabus scope for Embedded and IOT System Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: To Understand the IoT fundamentals and architecture modeling
- Official duration: As specified
- Official topic entries captured: 10
- The full source excerpt is preserved below for coverage checking.

▼ Introduction of IoT, History of IoT 1 Knowledge

Introduction of IoT, History of IoT 1 Knowledge is an official topic in Module 3 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Introduction of IoT, History of IoT 1 Knowledge using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, introduction of iot, history of iot 1 knowledge should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Introduction of IoT, History of IoT 1 Knowledge means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഇന്ററോഡക്ഷൻ ഓഫ് ഐടി, ഐടി ന്റെ ചരിത്രം 1
 Knowledge ഇന്ററോഡക്ഷൻ ഓഫ് ഐടി definition/meaning ഇന്ററോഡക്ഷൻ ഓഫ് ഐടി.
 ഇന്ററോഡക്ഷൻ ഓഫ് ഐടി, steps, application ഇന്ററോഡക്ഷൻ ഓഫ് ഐടി. Technical
 terms English-ഇന്ററോഡക്ഷൻ ഓഫ് ഐടി.

▼ IoT standards, IoT enabling technologies

IoT standards, IoT enabling technologies is an official topic in Module 3 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify IoT standards, IoT enabling technologies using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, IoT standards, IoT enabling technologies should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What IoT standards, IoT enabling technologies means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- IoT standards, IoT enabling technologies
 പട്ടികപ്പെടുത്തുക definition/meaning പട്ടികപ്പെടുത്തുക. പട്ടികപ്പെടുത്തുക
 പട്ടികപ്പെടുത്തുക, steps, application പട്ടികപ്പെടുത്തുക പട്ടികപ്പെടുത്തുക. Technical terms
 English- പട്ടികപ്പെടുത്തുക പട്ടികപ്പെടുത്തുക.

▼ IoT physical entities

IoT physical entities is an official topic in Module 3 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify IoT physical entities using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, IoT physical entities should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What IoT physical entities means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- IoT physical entities definition/meaning, steps, application. Technical terms English- Malayalam.

▼ IoT Building blocks - Architecture

IoT Building blocks - Architecture is an official topic in Module 3 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify IoT Building blocks - Architecture using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, iot building blocks - architecture should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What IoT Building blocks - Architecture means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- IoT Building blocks – Architecture
 പരമ്പരാഗതമായി നിർവ്വചിക്കുക definition/meaning നിർവ്വചിക്കുക. പരമ്പരാഗതമായി
 നിർവ്വചിക്കുക, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms
 English-ൽ നിർവ്വചിക്കുക നിർവ്വചിക്കുക.

▼ Study of existing IoT platforms

Study of existing IoT platforms is an official topic in Module 3 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Study of existing IoT platforms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, study of existing iot platforms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Study of existing IoT platforms means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓഡിയോ സ്റ്റഡി ഓഫ് ഇസ്റ്റിംഗ് ഐടി പ്ലാറ്റ്ഫോംസ്

ഓഡിയോ സ്റ്റഡി ഓഫ് ഐടി പ്ലാറ്റ്ഫോംസ് definition/meaning ഓഡിയോ സ്റ്റഡി ഓഫ് ഐടി പ്ലാറ്റ്ഫോംസ്. ഓഡിയോ സ്റ്റഡി ഓഫ് ഐടി പ്ലാറ്റ്ഫോംസ്, steps, application ഓഡിയോ സ്റ്റഡി ഓഫ് ഐടി പ്ലാറ്റ്ഫോംസ്. Technical terms English-ഓ ഓഡിയോ സ്റ്റഡി ഓഫ് ഐടി പ്ലാറ്റ്ഫോംസ്.

▼ Overview of Wireless sensor Networks 2 Knowledge

Overview of Wireless sensor Networks 2 Knowledge is an official topic in Module 3 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Overview of Wireless sensor Networks 2 Knowledge using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, overview of wireless sensor networks 2 knowledge should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Overview of Wireless sensor Networks 2 Knowledge means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Overview of Wireless sensor Networks 2 Knowledge definition/meaning . steps, application . Technical terms English- .

▼ Overview of RF, WiFi Bluetooth, Zigbee 2 Knowledge

Overview of RF, WiFi Bluetooth, Zigbee 2 Knowledge is an official topic in Module 3 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Overview of RF, WiFi Bluetooth, Zigbee 2 Knowledge using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, overview of rf, wifi bluetooth, zigbee 2 knowledge should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Overview of RF, WiFi Bluetooth, Zigbee 2 Knowledge means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Overview of RF, WiFi Bluetooth, Zigbee 2 Knowledge definition/meaning. Steps, application. Technical terms English- Malayalam.

▼ Overview of GSM/GPRS 2 Knowledge

Overview of GSM/GPRS 2 Knowledge is an official topic in Module 3 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Overview of GSM/GPRS 2 Knowledge using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, overview of gsm/gprs 2 knowledge should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Overview of GSM/GPRS 2 Knowledge means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Overview of GSM/GPRS 2 Knowledge
 പരമ്പരാഗതമായി നിർവ്വചന/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക
 നിർവ്വചനം, steps, application നിർവ്വചനം നിർവ്വചിക്കുക നിർവ്വചനം. Technical terms
 English- നിർവ്വചനം നിർവ്വചിക്കുക.

▼ Requirements of IoT, RFID Basics

Requirements of IoT, RFID Basics is an official topic in Module 3 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Requirements of IoT, RFID Basics using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, requirements of iot, rfid basics should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Requirements of IoT, RFID Basics means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഇന്റർനെറ്റ് ഓഫ് തിംഗ്സ് (IoT) യുടെ അടിസ്ഥാനപരമായ ആവശ്യകതകൾ, RFID അടിസ്ഥാനപരമായ ആവശ്യകതകൾ. ഇതിൽ ഉൾപ്പെട്ടിരിക്കുന്നവയെക്കുറിച്ച് നിങ്ങളുടെ പരീക്ഷയിൽ ഉപയോഗിക്കുക. നിങ്ങളുടെ ഉത്തരവ്, steps, application എന്നിവയെക്കുറിച്ച് വിശദമായി വിശദീകരിക്കുക. Technical terms English-ൽ ഉപയോഗിക്കുക.

▼ WSN topology and deployment strategies 2 Apply Introduction of IoT, History of IoT , IoT standards, IoT enabling technologies, IoT physical entities, IoT Building blocks - Architecture, Study of existing IoT platforms, Overview of Wireless sensor Networks ,Overview of RF, Overview of WiFi ,Bluetooth, Zigbee, Overview of GSM/ GPRS, Requirements of IoT ,RFID Basics ,WSN topology and deployment strategies To understand the smart things, functional blocks, communication networks

WSN topology and deployment strategies 2 Apply Introduction of IoT, History of IoT , IoT standards, IoT enabling technologies, IoT physical entities, IoT Building blocks - Architecture, Study of existing IoT platforms, Overview of Wireless sensor Networks ,Overview of RF, Overview of WiFi ,Bluetooth, Zigbee, Overview of GSM/ GPRS, Requirements of IoT ,RFID Basics ,WSN topology and deployment strategies To understand the smart things, functional blocks, communication networks is an official topic in Module 3 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify WSN topology and deployment strategies 2 Apply Introduction of IoT, History of IoT , IoT standards, IoT enabling technologies, IoT physical entities, IoT Building blocks - Architecture, Study of existing IoT platforms, Overview of Wireless sensor Networks ,Overview of RF, Overview of WiFi ,Bluetooth, Zigbee, Overview of GSM/ GPRS, Requirements of IoT ,RFID Basics ,WSN topology and deployment strategies To understand the

smart things, functional blocks, communication networks using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, wsn topology and deployment strategies 2 apply introduction of iot, history of iot , iot standards, iot enabling technologies, iot physical entities, iot building blocks – architecture, study of existing iot platforms, overview of wireless sensor networks ,overview of rf, overview of wifi ,bluetooth, zigbee, overview of gsm/ gprs, requirements of iot ,rfid basics ,wsn topology and deployment strategies to understand the smart things, functional blocks, communication networks should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What WSN topology and deployment strategies 2 Apply Introduction of IoT, History of IoT , IoT standards, IoT enabling technologies, IoT physical entities, IoT Building blocks – Architecture, Study of existing IoT platforms, Overview of Wireless sensor Networks ,Overview of RF, Overview of WiFi ,Bluetooth, Zigbee, Overview of GSM/ GPRS, Requirements of IoT ,RFID Basics ,WSN topology and deployment strategies To understand the smart things, functional blocks, communication networks means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-WSN topology and deployment strategies 2 Apply Introduction of IoT, History of IoT , IoT standards, IoT enabling technologies, IoT physical entities, IoT Building blocks – Architecture, Study of existing IoT platforms, Overview of Wireless sensor Networks ,Overview of RF, Overview of WiFi ,Bluetooth, Zigbee, Overview of GSM/ GPRS, Requirements of IoT ,RFID Basics ,WSN topology and deployment strategies To understand the smart things, functional blocks, communication networks പ്രവർത്തനം definition/meaning പ്രവർത്തനം . പ്രവർത്തനം പ്രവർത്തനം പ്രവർത്തനം , steps, application പ്രവർത്തനം പ്രവർത്തനം പ്രവർത്തനം . Technical terms English- പ്രവർത്തനം പ്രവർത്തനം .

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

and protocols used in IoT

This module is presented from the official syllabus scope for Embedded and IOT System Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and protocols used in IoT
- Official duration: As specified
- Official topic entries captured: 9
- The full source excerpt is preserved below for coverage checking.

▼ Introduction of M2M, Evolution of M2M 1 Knowledge

Introduction of M2M, Evolution of M2M 1 Knowledge is an official topic in Module 4 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Introduction of M2M, Evolution of M2M 1 Knowledge using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, introduction of m2m, evolution of m2m 1 knowledge should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Introduction of M2M, Evolution of M2M 1 Knowledge means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഇന്ററോഡക്ഷൻ ഓഫ് M2M, Evolution of M2M 1 Knowledge ഇന്ററോഡക്ഷൻ ഓഫ് definition/meaning ഇന്ററോഡക്ഷൻ. ഇന്ററോഡക്ഷൻ ഇന്ററോഡക്ഷൻ, steps, application ഇന്ററോഡക്ഷൻ ഇന്ററോഡക്ഷൻ ഇന്ററോഡക്ഷൻ. Technical terms English-ഇന്ററോഡക്ഷൻ ഇന്ററോഡക്ഷൻ.

▼ Working of M2M, Role of M2M in IoT 2 Understanding IoT communication pattern, Standardization activities

Working of M2M, Role of M2M in IoT 2 Understanding IoT communication pattern, Standardization activities is an official topic in Module 4 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Working of M2M, Role of M2M in IoT 2 Understanding IoT communication pattern, Standardization activities using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, working of m2m, role of m2m in iot 2 understanding iot communication pattern, standardization activities should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Working of M2M, Role of M2M in IoT 2 Understanding IoT communication pattern, Standardization activities means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Working of M2M, Role of M2M in IoT 2 Understanding IoT communication pattern, Standardization activities □□□□□□□□□□ □□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□.

▼ 2 Understanding towards protocols

2 Understanding towards protocols is an official topic in Module 4 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding towards protocols using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding towards protocols should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding towards protocols means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 2 Understanding towards protocols
 ഫലപ്രസാദം definition/meaning. ഫലപ്രസാദം ഫലപ്രസാദം
 ഫലപ്രസാദം, steps, application ഫലപ്രസാദം ഫലപ്രസാദം. Technical terms
 English- ഫലപ്രസാദം ഫലപ്രസാദം.

▼ TCP/IP Basics, Overview of IPV6 2 Knowledge

TCP/IP Basics, Overview of IPV6 2 Knowledge is an official topic in Module 4 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify TCP/IP Basics, Overview of IPV6 2 Knowledge using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, tcp/ip basics, overview of ipv6 2 knowledge should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What TCP/IP Basics, Overview of IPV6 2 Knowledge means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- TCP/IP Basics, Overview of IPV6 2
 Knowledge definition/meaning .
 , steps, application . Technical
 terms English- .

▼ Network devices & configurations

Network devices & configurations is an official topic in Module 4 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Network devices & configurations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, network devices & configurations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Network devices & configurations means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-നൂറ് Network devices & configurations
 നൂറ് definition/meaning നൂറ്. നൂറ്
 നൂറ്, steps, application നൂറ് നൂറ്. Technical terms
 English-നൂറ് നൂറ്.

▼ Overview of Web servers , Socket programming 2 Apply

Overview of Web servers , Socket programming 2 Apply is an official topic in Module 4 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Overview of Web servers , Socket programming 2 Apply using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, overview of web servers , socket programming 2 apply should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Overview of Web servers , Socket programming 2 Apply means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
Overview of Web servers , Socket programming 2 Apply definition/meaning
steps, application
Technical terms English-

▼ IoT Protocol overview 1 Knowledge

IoT Protocol overview 1 Knowledge is an official topic in Module 4 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify IoT Protocol overview 1 Knowledge using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, iot protocol overview 1 knowledge should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What IoT Protocol overview 1 Knowledge means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- IoT Protocol overview 1 Knowledge
 definition/meaning. , steps, application . Technical terms
 English- .

▼ Overview of MQTT ,Overview of COAP,6lowpan

Overview of MQTT ,Overview of COAP,6lowpan is an official topic in Module 4 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Overview of MQTT ,Overview of COAP,6lowpan using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, overview of mqtt ,overview of coap,6lowpan should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Overview of MQTT ,Overview of COAP,6lowpan means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Overview of MQTT ,Overview of COAP,6lowpan definition/meaning .
 , steps, application .
 Technical terms English- .

▼ **Overview of REST, HTTP/HTTPS 1 Understanding Introduction of M2M, Evolution of M2M ,Working of M2M, Role of M2M in IoT, IoT communication pattern, Standardization activities towards protocols, TCP/IP Basics, Overview of IPV6, Network devices & configurations, Overview of Web servers , Socket programming, IoT Protocol overview, Overview of MQTT ,Overview of COAP,6lowpan, Overview of REST, HTTP/HTTPS. Text / Reference T/R Book Title/Author T1 Introduction to Embedded Systems by K V Shibu , 2nd edition, McGraw Hill, 2016 Embedded Systems - Architecture Programming and Design by Raj Kamal, 2nd T2 edition, McGraw Hill Muhammed Ali Mazidi & Janice Gilli Mazidi, R.D. Kinley, The 8051 T3 microcontroller and Embedded System, Pearson Education, 2nd edition. Vijay Madisetti and ArshdeepBahga, “Internet of Things (A Hands-on- T4 Approach)”, 1stEdition, VPT, 2014 David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome T5 Henry, —IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, Cisco Press, 2017 Rajkamal, “Internet of Things : Architecture and Design Principles”, McGraw Hill T6 (India) Private Limited R1 Embedded Real Time Systems by Dr.K.V.K.K Prasad Dream tech press, 2009. R2 The 8051 Microcontrollers: Architecture Programming and Applications, K Uma Rao & Andhe Pallavi, Pearson, 2011. R3 Olivier Hersent, David Boswarthick, Omar Elloumi , “The Internet of Things -Key applications and Protocols”, Wiley, 2012 Al-Fuqaha et al. Internet of things: A survey on enabling technologies, protocols, R4 and applications. IEEE Communications Surveys & Tutorials (2015), pp. 2347- 2376. Online Resources Sl.No Website Link <https://nptel.ac.in/courses/108102169> 1 Introduction to Embedded System Design, IIT Jammu <https://nptel.ac.in/courses/117104072> 2 Microcontrollers and applications, IIT Kharagpur <https://nptel.ac.in/courses/106105166> Introduction to internet of things, II 3 Introduction to Internet of Things, IIT Kharagpur K**

Overview of REST, HTTP/HTTPS 1 Understanding Introduction of M2M, Evolution of M2M ,Working of M2M, Role of M2M in IoT, IoT communication pattern, Standardization activities towards protocols, TCP/IP Basics,

Overview of IPV6, Network devices & configurations, Overview of Web servers , Socket programming, IoT Protocol overview, Overview of MQTT ,Overview of COAP,6lowpan, Overview of REST, HTTP/HTTPS. Text / Reference T/R Book Title/Author T1 Introduction to Embedded Systems by K V Shibu , 2nd edition, McGraw Hill, 2016 Embedded Systems – Architecture Programming and Design by Raj Kamal, 2nd T2 edition, McGraw Hill Muhammed Ali Mazidi & Janice Gilli Mazidi, R.D. Kinley, The 8051 T3 microcontroller and Embedded System, Pearson Education, 2nd edition. Vijay Madisetti and ArshdeepBahga, “Internet of Things (A Hands-on- T4 Approach)”,1stEdition, VPT, 2014 David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome T5 Henry, —IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, Cisco Press, 2017 Rajkamal, “Internet of Things : Architecture and Design Principles”, McGraw Hill T6 (India) Private Limited R1 Embedded Real Time Systems by Dr.K.V.K.K Prasad Dream tech press, 2009. R2 The 8051 Microcontrollers: Architecture Programming and Applications, K Uma Rao & Andhe Pallavi, Pearson, 2011. R3 Olivier Hersent, David Boswarthick, Omar Elloumi , “The Internet of Things –Key applications and Protocols”, Wiley, 2012 Al-Fuqaha et al. Internet of things: A survey on enabling technologies, protocols, R4 and applications. IEEE Communications Surveys & Tutorials (2015), pp. 2347- 2376. Online Resources Sl.No Website Link
<https://nptel.ac.in/courses/108102169> 1 Introduction to Embedded System Design, IIT Jammu <https://nptel.ac.in/courses/117104072> 2 Microcontrollers and applications, IIT Kharagpur <https://nptel.ac.in/courses/106105166> Introduction to internet of things, II 3 Introduction to Internet of Things, IIT Kharagpur K is an official topic in Module 4 of Embedded and IOT System Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official

source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Overview of REST, HTTP/HTTPS 1 Understanding Introduction of M2M, Evolution of M2M ,Working of M2M, Role of M2M in IoT, IoT communication pattern, Standardization activities towards protocols, TCP/IP Basics, Overview of IPV6, Network devices & configurations, Overview of Web servers , Socket programming, IoT Protocol overview, Overview of MQTT ,Overview of COAP,6lowpan, Overview of REST, HTTP/HTTPS. Text / Reference T/R Book Title/Author T1 Introduction to Embedded Systems by K V Shibu , 2nd edition, McGraw Hill, 2016 Embedded Systems – Architecture Programming and Design by Raj Kamal, 2nd T2 edition, McGraw Hill Muhammed Ali Mazidi & Janice Gilli Mazidi, R.D. Kinley, The 8051 T3 microcontroller and Embedded System, Pearson Education, 2nd edition. Vijay Madiseti and ArshdeepBahga, “Internet of Things (A Hands-on- T4 Approach)”,1stEdition, VPT, 2014 David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome T5 Henry, —IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, Cisco Press, 2017 Rajkamal, “Internet of Things : Architecture and Design Principles”, McGraw Hill T6 (India) Private Limited R1 Embedded Real Time Systems by Dr.K.V.K.K Prasad Dream tech press, 2009. R2 The 8051 Microcontrollers: Architecture Programming and Applications, K Uma Rao & Andhe Pallavi, Pearson, 2011. R3 Olivier Hersent, David Boswarthick, Omar Elloumi , “The Internet of Things –Key applications and Protocols”, Wiley, 2012 Al-Fuqaha et al. Internet of things: A survey on enabling technologies, protocols, R4 and applications. IEEE Communications Surveys & Tutorials (2015), pp. 2347- 2376. Online Resources SI.No Website Link <https://nptel.ac.in/courses/108102169> 1 Introduction to Embedded System Design, IIT Jammu <https://nptel.ac.in/courses/117104072> 2 Microcontrollers and applications, IIT Kharagpur <https://nptel.ac.in/courses/106105166>Introduction to internet of things, II 3 Introduction to Internet of Things, IIT Kharagpur K using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, overview of rest, http/https 1 understanding introduction of m2m, evolution of m2m ,working of m2m, role of m2m in iot, iot communication pattern, standardization activities towards protocols, tcp/ip basics, overview of ipv6, network devices & configurations, overview of web servers , socket programming, iot protocol overview, overview of mqtt ,overview of coap,6lowpan, overview of rest, http/https. text / reference t/r book title/author t1 introduction to embedded systems by k v shibu , 2nd edition, mcgraw hill, 2016 embedded systems – architecture programming and design by raj kamal, 2nd t2 edition, mcgraw hill muhammed ali mazidi & janice gilli mazidi, r.d. kinley, the 8051 t3 microcontroller and embedded system, pearson education, 2nd edition. vijay madiseti and arshdeepbahga, “internet of things (a hands-on- t4 approach)”,1stedition, vpt, 2014 david hanes, gonzalo salgueiro, patrick grossetete, rob barton and jerome t5 henry, —iot fundamentals: networking technologies, protocols and use cases for internet of things, cisco press, 2017 rajkamal, “internet of things : architecture and design principles”, mcgraw hill t6 (india) private limited r1 embedded real time systems by dr.k.v.k.k prasad dream tech press, 2009. r2 the 8051 microcontrollers: architecture programming and applications, k uma rao & andhe pallavi, pearson, 2011. r3 olivier hersent, david boswarthick, omar elloumi , “the internet of things –key applications and protocols”, wiley, 2012 al-fuqaha et al. internet of things: a survey on enabling technologies, protocols, r4 and applications. ieee communications surveys & tutorials (2015), pp. 2347- 2376. online resources sl.no website link <https://nptel.ac.in/courses/108102169> 1 introduction to embedded system design, iit jammu <https://nptel.ac.in/courses/117104072> 2 microcontrollers and applications, iit kharagpur <https://nptel.ac.in/courses/106105166> introduction to internet of things, ii 3 introduction to internet of things, iit kharagpur k should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Overview of REST, HTTP/HTTPS 1 Understanding Introduction of M2M, Evolution of M2M ,Working of M2M, Role of M2M in IoT, IoT communication pattern, Standardization activities towards protocols, TCP/IP Basics, Overview of IPV6, Network devices & configurations, Overview of Web servers , Socket programming, IoT Protocol overview, Overview of MQTT ,Overview of COAP,6lowpan, Overview of REST, HTTP/HTTPS. Text / Reference T/R Book Title/Author T1 Introduction to Embedded Systems by K V Shibu , 2nd edition, McGraw Hill, 2016 Embedded Systems – Architecture Programming and Design by Raj Kamal, 2nd T2 edition, McGraw Hill Muhammed Ali Mazidi & Janice Gilli Mazidi, R.D. Kinley, The 8051 T3 microcontroller and Embedded System, Pearson Education, 2nd edition. Vijay Madisetti and ArshdeepBahga, “Internet of Things (A Hands-on- T4 Approach)”,1stEdition, VPT, 2014 David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome T5 Henry, —IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, Cisco Press, 2017 Rajkamal, “Internet of Things : Architecture and Design Principles”, McGraw Hill T6 (India) Private Limited R1 Embedded Real Time Systems by Dr.K.V.K.K Prasad Dream tech press, 2009. R2 The 8051 Microcontrollers: Architecture Programming and Applications, K Uma Rao & Andhe Pallavi, Pearson, 2011. R3 Olivier Hersent, David Boswarthick, Omar Elloumi , “The Internet of Things –Key applications and Protocols”, Wiley, 2012 Al-Fuqaha et al. Internet of things: A survey on enabling technologies, protocols, R4 and applications. IEEE Communications Surveys & Tutorials (2015), pp. 2347- 2376. Online Resources SI.No Website Link https://nptel.ac.in/courses/108102169 1 Introduction to Embedded System Design, IIT Jammu https://nptel.ac.in/courses/117104072 2 Microcontrollers and applications, IIT Kharagpur https://nptel.ac.in/courses/106105166Introduction to internet of things, II 3 Introduction to Internet of Things, IIT Kharagpur K means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally

supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Overview of REST, HTTP/HTTPS 1 Understanding Introduction of M2M, Evolution of M2M ,Working of M2M, Role of M2M in IoT, IoT communication pattern, Standardization activities towards protocols, TCP/IP Basics, Overview of IPV6, Network devices & configurations, Overview of Web servers , Socket programming, IoT Protocol overview, Overview of MQTT ,Overview of COAP,6lowpan, Overview of REST, HTTP/HTTPS. Text / Reference T/R Book Title/Author T1 Introduction to Embedded Systems by K V Shibu , 2nd edition, McGraw Hill, 2016 Embedded Systems – Architecture Programming and Design by Raj Kamal, 2nd T2 edition, McGraw Hill Muhammed Ali Mazidi & Janice Gilli Mazidi, R.D. Kinley, The 8051 T3 microcontroller and Embedded System, Pearson Education, 2nd edition. Vijay Madisetti and ArshdeepBahga, “Internet of Things (A Hands-on- T4 Approach)”,1stEdition, VPT, 2014 David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome T5 Henry, —IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, Cisco Press, 2017 Rajkamal, “Internet of Things : Architecture and Design Principles”, McGraw Hill T6 (India) Private Limited R1 Embedded Real Time Systems by Dr.K.V.K.K Prasad Dream tech press, 2009. R2 The 8051 Microcontrollers: Architecture Programming and Applications, K Uma Rao & Andhe Pallavi, Pearson, 2011. R3 Olivier Hersent, David Boswarthick, Omar Elloumi , “The Internet of Things –Key applications and Protocols”, Wiley, 2012 Al-Fuqaha et al. Internet of things: A survey on enabling technologies, protocols, R4 and applications. IEEE Communications Surveys & Tutorials (2015), pp. 2347- 2376. Online Resources Sl.No Website Link
https://nptel.ac.in/courses/108102169 1 Introduction to Embedded System Design, IIT Jammu https://nptel.ac.in/courses/117104072 2 Microcontrollers and applications, IIT Kharagpur
https://nptel.ac.in/courses/106105166Introduction to internet of things, II 3 Introduction to Internet of Things, IIT Kharagpur K definition/meaning steps,

application ഏകദേശം അളക്കുന്നതിനായി ഉപയോഗിക്കുന്നു. Technical terms English-ൽ ഉപയോഗിക്കുന്ന പദങ്ങൾ.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

- M4.03 2 Understanding towards protocols
- M4.04 TCP/IP Basics, Overview of IPV6 2 Knowledge
- M4.05 Network devices & configurations 1 Understanding
- M4.06 Overview of Web servers , Socket programming 2 Apply
- M4.07 IoT Protocol overview 1 Knowledge
- M4.08 Overview of MQTT ,Overview of COAP,6lowpan 2 Understanding
- M4.09 Overview of REST, HTTP/HTTPS 1 Understanding
- Series Test - II 1
- Contents:
- Introduction of M2M, Evolution of M2M ,Working of M2M, Role of M2M in IoT, IoT
- communication pattern, Standardization activities towards protocols, TCP/IP Basics,
- Overview of IPV6, Network devices & configurations, Overview of Web servers , Socket
- programming, IoT Protocol overview, Overview of MQTT ,Overview of COAP,6lowpan,
- Overview of REST, HTTP/HTTPS. Text / Reference
- T/R Book Title/Author
- T1 Introduction to Embedded Systems by K V Shibu , 2nd edition, McGraw Hill, 2016
- Embedded Systems - Architecture Programming and Design by Raj Kamal, 2nd edition, McGraw Hill
- Muhammed Ali Mazidi & Janice Gilli Mazidi, R.D. Kinley, The 8051
- microcontroller and Embedded System, Pearson Education, 2nd edition.
- Vijay Madiseti and ArshdeepBahga, "Internet of Things (A Hands-on-Approach)",1stEdition, VPT, 2014

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain History, Embedded Systems vs general computing 1 Knowledge systems Describe the Characteristics and quality attributes of. (3 marks)**

► **Define or explain 2 Understanding Embedded systems,. (3 marks)**

► **Define or explain Explain the Classification of Embedded systems 1 Understanding Describe about the Core of a typical embedded. (3 marks)**

► **Define or explain systems, Memory, Sensor and actuators, 3 Understanding Communication Interface, Embedded firmware Explain the Major application areas of embedded. (3 marks)**

Module 2

► **Define or explain Programming Embedded Systems- Processor types 1 Understanding. Explain about Selection of programming language,. (3 marks)**

► **Define or explain 2 Understanding Selection of Operating System Describe about the Development of Embedded System. (3 marks)**

► **Define or explain 2 Understanding. Software Block diagram of 8051,**

Pin configuration, Registers,. (3 marks)

- ▶ **Define or explain 2 Understanding. Internal Memory, Timers: Modes and Applications, Port Structures,. (3 marks)**

Module 3

- ▶ **Define or explain Introduction of IoT, History of IoT 1 Knowledge. (3 marks)**

- ▶ **Define or explain IoT standards, IoT enabling technologies. (3 marks)**

- ▶ **Define or explain IoT physical entities. (3 marks)**

- ▶ **Define or explain IoT Building blocks - Architecture. (3 marks)**

Module 4

- ▶ **Define or explain Introduction of M2M, Evolution of M2M 1 Knowledge. (3 marks)**

- ▶ **Define or explain Working of M2M, Role of M2M in IoT 2 Understanding IoT communication pattern, Standardization activities. (3 marks)**

- ▶ **Define or explain 2 Understanding towards protocols. (3 marks)**

- ▶ **Define or explain TCP/IP Basics, Overview of IPV6 2 Knowledge. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **History, Embedded Systems vs general computing 1 Knowledge systems Describe the Characteristics and quality attributes of.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Programming Embedded Systems- Processor types 1 Understanding. Explain about Selection of programming language,.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Introduction of IoT, History of IoT 1 Knowledge.**

Module 4

1-mark definition: State the meaning of **Introduction of M2M, Evolution of M2M 1 Knowledge.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://nptel.ac.in/courses/108102169>
- Introduction to Embedded System Design, IIT Jammu
<https://nptel.ac.in/courses/117104072>
- Microcontrollers and applications, IIT Kharagpur
- <https://nptel.ac.in/courses/106105166> Introduction to internet of things, II
- Introduction to Internet of Things, IIT Kharagpur K

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5303C. Verify institutional updates against the current official curriculum.